

Day 5 Case Study

1. A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

Check the power adapter: Make sure the charger is properly connected and check for any damage to the cable or adapter.

Check the power outlet: Plug another device into the outlet to confirm that it is supplying power.

Check the charging port: Inspect the laptop's charging port for loose connections, dust, or physical damage.

Perform a power reset: Disconnect the charger, remove the battery if possible, hold the power button for 15–20 seconds, then reconnect the charger and try powering on.

Check for hardware signs: Look for charging LEDs, fan movement, or any sound when pressing the power button to determine whether the laptop is receiving power or has an internal hardware problem.

2. You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.

1. **Check the monitor:** Make sure the monitor is powered on and the power indicator light is working. Also confirm the correct input source is selected.
2. **Check display cables:** Disconnect and reconnect the HDMI, DisplayPort, or VGA cable at both ends. Check the cable for visible damage and try another cable if available.
3. **Check the PC connection:** Make sure the display cable is connected to the correct port, especially if the PC has a dedicated graphics card. Ensure the GPU is properly seated if necessary.
4. **Check display settings:** Try pressing **Windows + P** and select **Duplicate** or **PC screen only**. If possible, check the display resolution and refresh rate.
5. **Test another monitor:** Connect the PC to a different working monitor or TV to determine whether the problem is with the monitor or the computer.
6. **Restart and confirm the fix:** Restart the PC after making the changes and check whether the display returns to normal. If the screen is still blank, the GPU, monitor, or another hardware component may need further testing.

3. A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

First, I would check the **mouse itself** by replacing its batteries, turning it off and on, and checking whether its wireless receiver is connected properly. I would then plug the receiver into a **different USB port** to see if the mouse starts working. If it works in another port, the original USB port may be faulty.

Next, I would test the mouse on **another computer**. If it still does not work, the mouse or receiver is likely faulty. If it works on another computer, I would check the original system's **Bluetooth/USB settings and Device Manager** for disabled devices or driver problems. After updating or reinstalling the relevant driver, I would reconnect the mouse and confirm that it responds normally.

4. Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.

So here are three possible reasons

1. **Faulty or blocked fans:** Check whether the CPU and case fans are spinning properly and confirm that no cables are blocking them. Replace any fan that is not working.
2. **Dust buildup:** Inspect the CPU cooler, GPU, fans, and air vents for dust. Clean the components carefully with compressed air and make sure airflow is not blocked.
3. **Dried thermal paste:** Check the thermal paste between the CPU and cooler. If it is dried or damaged, clean the old paste and apply a fresh, suitable layer before reinstalling the cooler.